



Ayush Yadav

Roll No.:240102124

B.Tech. - Electronics & Communication Engineering

Minor in Computer Science and Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology Guwahati	8.40 (Current)	2024-Present
B.Tech. Minor	Indian Institute of Technology Guwahati	9 (Current)	2025-Present
Senior Secondary	CBSE Board	86.8%	2023
Secondary	CBSE Board	97.6%	2021

PROJECTS

- **Keus - Bot Design Challenge** Feb. 2026 - March 2026
Industrial PS offered by Keus during Kriti - Inter Hostel Tech Competition, 2026 Github
 - Engineered an **Arduino** based autonomous **line following bot** using an **IR sensor array** and **PID-tuned control algorithms** for high-speed line following and trajectory correction.
 - Integrated **dynamic obstacle detection and avoidance** using Ultrasonic sensors and a **Raspberry Pi** vision pipeline to enable **real-time image detection** and low-latency live camera feed.
 - Developed a **Wi-Fi-enabled web dashboard** for real-time telemetry, displaying identified object name and mission timer.
- **Synaptron- EMG based Gesture Recognition System** Oct. 2025 - Present
Personal Project Drive
 - Designed an **Arduino-based prosthetic system**, translating Surface EMG muscle activations into real time, **full-hand grip maneuvers** through optimized sensor interfacing and responsive motor control.
 - Formulated a robust **signal conditioning pipeline** incorporating **FIR band-pass filtering, rectification, and smoothing** to extract stable control signals from noisy surface EMG data for reliable prosthetic actuation.
 - Synchronized **multi-channel PWM servo logic** with **optimized ADC sampling** rates to virtually eliminate control loop latency while substantially enhancing real-time grip stability and finger positioning accuracy.
- **Rain Rain Go Away- Smart Rain-Sensing Wiper Control System** Feb. 2025 - Mar. 2025
Kriti - Inter Hostel Tech Competition, 2025 Github
 - Synthesized a **rain-intensity-based wiper system** featuring a custom **Flash ADC** built with **LM393 comparators** and a precision resistor ladder for real-time rain level classification.
 - Streamlined power delivery using a **Buck converter** and **PMOS-based voltage selection logic**, enabling efficient DC motor speed control while minimizing overall power dissipation.
 - Made a **bidirectional motion** control unit utilizing **JK flip-flop sequential logic** and **Hall-effect sensors** to achieve automated, position-based wiper reversal via an H-Bridge.
- **Live DC Circuit Simulator** Nov. 2025
Course Project Github
 - Designed high-performance **interactive circuit simulator** that enables users to architect custom schematics and perform **real-time visualization** of voltage transients and current flow across all RLC components.
 - Applied **Modified Nodal Analysis (MNA)** to compute node voltages and branch currents for accurate circuit analysis.
 - Built a computation engine and GUI for **user-defined RLC circuits** with **dynamic parameter updates** and live visualization.

TECHNICAL SKILLS

- **Digital Electronics:** Verilog, Xilinx Vivado
- **Electronics:** LTSpice*, MATLAB*
- **Programming:** C++, C, Python, Data Structure & Algorithms, OOP*, Git*
- **Mathematics:** Linear Algebra, Real Analysis, Complex Analysis, Probability

* Elementary proficiency

KEY COURSES TAKEN

- **Electronics:** Digital Circuits (Theory + Lab), Analog Circuits (Theory + Lab), Circuit Theory, Control Systems, Signals and Systems, Semiconductor Devices, Measurement and Instrumentation (Theory + Lab).
- **Mathematics:** Linear Algebra, Single and Multi-Variable Calculus, Probability & Random Processes.
- **Computer Science:** Introduction to Computing (Theory + Lab), Theoretical Foundations of Computer Science, Data Structures and Algorithms.

ACHIEVEMENTS & EXTRACURRICULARS

- **Kriti '25**, Won **Gold Medal** in **Electronics** during Kriti, IIT Guwahati's **Inter-Hostel Tech Competition.** 2025
- **Kriti '26**, Won **Silver Medal** in the **Bot Design Challenge** offered by Keus during Kriti 2026. 2026
- **Academic Merit**, Earned merit-based **Branch Change** to Electronics and Communication Engineering 2025
- **Sports**, Won **Gold and Bronze Medal** in Hockey and Kho-Kho during Spardha, Inter-Hostel Sports Competition. 2025
- **IICM**, Secured **Rank 5** in Group Dance during Inter IIT Cultural Meet 8.0, IIT Kanpur. 2025
- **Dance**, Won **Gold Medal** in Group Dance during Manthan'25, IIT Guwahati's Inter-Hostel Cultural Competition. 2025
- **Sketch Video**, Secured **First Position** in Quite Sketchy during Manthan'25. 2025